LUNARSAFE // OPERATIONAL CAPABILITIES, LIMITATIONS & FLIGHT RISKS

MISSION READINESS BOUNDARIES · SENSOR ASSUMPTIONS · ETHICAL AUDIT

STANDARD: ISRO Safety Margins · Planetary Protection · Flight Certification Criteria

SECTION 1 — SYSTEM CAPABILITIES (WHAT LUNARSAFE CAN DO)

Deterministic DEM Slope	Computes exact physical slopes [0° - 90°] from IEEE 754 Float32 DEMs with zero stochastic variation.
First-Order Error Propagation	Calculates analytical per-pixel slope uncertainty based on sensor vertical accuracy (sigma_z = 1.0m).
ISRO 24m Conservative Filtering	Applies a 95.45% CI (mean + 2*sigma <= 10.0°) safe cell classification per Chandrayaan standards.
Automated Hazard Extraction	Detects craters and boulders exceeding threshold hazard depths (1.0m) and heights (0.30m).
Cryptographic Evidence Chain	Seals all pipeline steps into SHA-256 digests with Ed25519 signatures and AES-256-GCM encryption.

SECTION 2 — TECHNICAL LIMITATIONS (WHAT LUNARSAFE CANNOT DO)

1. Subsurface Hazard Invisibility: LunarSafe operates strictly on optical surface reflectance and topocentric elevation. It cannot detect buried boulders, subsurface lava tubes, or porous dust pockets with insufficient load-bearing capacity.

2. Geotechnical Soil Mechanics: The system does not model regolith shear strength, cohesion, or sinkage dynamics under lander footpads.

3. Dynamic Thruster Plume Interaction: The engine does not simulate regolith erosion, optical blackout, or dust blowing caused by terminal descent thruster exhaust.

4. Monochromatic Optical Illumination Dependence: When operating on optical imagery without DEM, slope estimates rely on calibrated gradients which assume uniform regolith albedo and steady illumination.

SECTION 3 — CORE SENSOR & PHYSICAL ASSUMPTIONS

Panchromatic Spectrum	Input imagery must be single-band panchromatic lunar reflectance (chromatic variance < 0.1%).
Orthorectified Geometry	Input rasters must be geometrically rectified to true ground sample distance (GSD meters/pixel).
Independent Elevation Errors	Uncertainty propagation assumes i.i.d. Gaussian error distribution per laser altimeter pulse.
Local Planar Topography	Finite difference stencils assume terrain behaves as locally differentiable planar facets over 3x3 windows.

SECTION 4 — FLIGHT DEPENDENCIES & GROUND STATUS

LunarSafe has been rigorously validated against genuine NASA LOLA DEMs and ISRO Chandrayaan-2 orbital products. However, it currently lacks in-flight validation aboard an active lunar descent vehicle. Prior to mission deployment, hardware-in-the-loop (HIL) testing on space-qualified radiation-hardened DSPs/FPGAs (e.g., UT699 or Virtex-5QV) must be executed.

SECTION 5 — MISSION RISKS & APPLIED MITIGATIONS

Risk: Undetected Boulder in Shadow	Mitigation: 2-sigma conservative uncertainty culling filters out high-variance shaded cells.
Risk: Division by Zero in Grid Sizing	Mitigation: Explicit assertions prevent GSD > 24m from triggering invalid grid partitioning.
Risk: Code Tampering in Flight	Mitigation: Frozen engine SHA-256 seal (8CA34C...2882956) guarantees bit-identical immutability.

SECTION 6 — ETHICAL & SCIENTIFIC INTEGRITY CONSIDERATIONS

In accordance with COSPAR Planetary Protection guidelines and open scientific standards, LunarSafe guarantees absolute honesty in all reported errors. Scientific fraud (such as truncating or altering official ISRO PDS4 rasters or suppressing unfavorable validation metrics) is strictly prohibited. All validation data, formulas, and test suites are fully open, reproducible, and verifiable.